

An Intriguing Networking Choice: Passive Optical LAN (POL)

一个耐人寻味的选择：无源光局域网（POL）

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正文 / Body

The networking journey is always one of change.

网络的发展历程永远伴随着变化。

Yet there are some technologies that we've used for literally decades that are showing signs of being less than up to the challenges of growing bandwidth and device density.

然而，有些我们已经使用了数十年的技术，正逐渐显示出难以应对不断增长的带宽和设备密度。

Indeed, new services such as Wi-Fi 6 are driving the need for more bandwidth, especially on the backhaul.

事实上，Wi-Fi 6 等新业务正在推动对更高带宽的需求，尤其是在回传（backhaul）侧。

Copper Ethernet cables are a great example of where a great technology is beginning to run into the harsh limits of physics.

铜缆以太网就是一个很好的例子：一项伟大的技术正开始撞上物理规律的天花板。

To achieve even 10 Gigabit speeds excessive care has to be taken with thick, less flexible cables and distance limits of 100 meters for properly installed cabling systems (Cat6a or Cat7).

即便只是想达到 10 Gb 速率，也必须对粗硬、不易弯折的线缆格外小心；而且规范安装的布线系统（Cat6a 或 Cat7）有 **100 米** 的距离限制。

With bandwidth rising to 40Gbps and 100Gbps and beyond, it becomes a matter of much discussion on how we support networks going forward, especially in new construction.

当带宽迈向 40Gbps、100Gbps 乃至更高，如何支撑未来的网络——尤其是在新建建筑中——就成了一个值得反复讨论的问题。

Nobody wants the expense of upgrading a building's network cabling within a few years of finishing it.

没有人愿意在一栋楼刚建成没几年，就再花钱升级它的网络布线。

One of the interesting alternatives out there is called POL or Passive Optical LAN.

有意思的替代方案之一叫做 POL，即无源光局域网（Passive Optical LAN）。

POL is a subset of optical networking technology under the greater umbrella of PON (Passive Optical Network).

POL 是更大的 **PON（无源光网络，Passive Optical Network）** 技术伞下的一个子集。

These technologies are tested and have been in use for a long time, mostly in Fiber to the Home installations, metro installations, and large campus installations.

这些技术经过充分验证、已被使用很久，主要用于光纤到户（FTTH）部署、城域部署和大型院区部署。

We are not going to go into all the variations of PON, there are a bewildering array, but confine the discussion to POL and enterprise use cases.

我们不打算展开 PON 的全部变种——它们多得让人眼花缭乱——而只讨论 POL 与企业场景。

POL itself has been around for quite a while too, but never really took off, but there is reason to believe that now POL could become a great alternative to traditional designs for enterprises.

POL 本身也已存在相当长时间，但一直没有真正流行起来；不过现在有理由相信，POL 可能成为企业传统设计的一个优秀替代方案。

Let's dive down a little bit into what we mean by POL.

我们来稍微深入看看 POL 到底指什么。

POL uses passive optical cabling, single mode fiber.

POL 使用无源的光缆，即**单模光纤**。

Passive means that only the source and receiver require electrical power.

无源（passive）的意思是：**只有光源（发送端）和接收端需要电力**。

The optical splitters in between are unpowered, which makes them less expensive, less prone to failure, and do not require battery backups.

中间的光分路器不供电，这使它们更便宜、更不易故障，也不需要后备电池。

Plus, they can save considerable power vs. active solutions.

此外，相比有源方案，它们还能节省可观的电力。

POL networks can be used for several purposes, such as carrying data, video, and phone services on separate strands of the same optical cable.

POL 网络可用于多种用途，比如在同一根光缆的不同纤芯上分别承载数据、视频和电话业务。

The first question most people ask when they understand POL is a simple "why" while looking at the optical links available on campus switches.

大多数人在理解 POL 后问的第一个问题都很简单——“为什么？”，同时看着院区交换机上已有的那些光口。

The answer to that comes down to a simpler optical system, upgradability and reliability.

答案归结为：更简单的光系统、可升级性和可靠性。

POL networks do not have the failures associated with pluggable optical transceivers, which are both expensive and fragile.

POL 网络没有可插拔光模块（pluggable optical transceivers）带来的那些故障——它们既昂贵又脆弱。

Lastly, the POL network is upgradable in place far in excess of what can be done with copper, with new speeds available with minimal costs compared to traditional networking designs.

最后，POL 网络的**原地升级**能力远超铜缆：相比传统网络设计，能以极小的成本获得新的速率。

There is more as well.

还有更多好处。

In marine or marine adjacent environments, POL offers a non-corroding alternative that is impervious to the salt air.

在海洋或近海环境中，POL 提供了一种不会腐蚀、不受盐雾影响的替代方案。

Plus, POL can be used at distances up to 20km.

此外，POL 可用于最远 **20 公里** 的距离。

This makes POL a great choice for if you are administering a campus for connecting buildings, setting up video security, and connecting campus-wide Wi-Fi.

这使 POL 成为管理院区时的绝佳选择：连接楼宇、部署视频安防、连接覆盖全院区的 Wi-Fi。

POL makes these functions easier to both install and to maintain long term.

POL 让这些功能在安装和长期维护上都更容易。

Companies are also increasingly aware of operational efficiencies.

企业也越来越意识到运营效率。

POL networking, due to its simple nature is much easier to administer, even on a large campus.

由于结构简单，POL 网络即使在大院区里也更容易管理。

Companies considering POL should select a vendor that either sells both the POL and Ethernet solution and has management under a single pane of glass or select a POL vendor that has done the necessary integration with your existing Ethernet networking solution.

考虑 POL 的企业应当选择这样的厂商：要么同时提供 POL 和以太网方案、且能在单一管理界面（single pane of glass）下统一管理；要么选择已经与你现有以太网方案完成必要集成的 POL 厂商。

The ripest targets for POL are generally new construction and historical structures.

POL 最适合的目标通常是**新建建筑和历史建筑**。

For new construction, planning and installing an optical fiber network that should have a functional life of decades is an easy choice.

对新建建筑来说，规划并安装一套使用寿命可达数十年的光纤网络是个轻松的选择。

For historical structures where there can be no or extremely minimal changes to the structure, POL offers a long life and fairly small cabling to minimize the compromises that need to be taken.

对历史建筑，要求结构上几乎不能改动，POL 长寿命、线缆细小的优势能把妥协降到最低。

Existing brownfield environments can be retrofitted for POL, but often the costs of installing optical cabling in spaces not designed for optical can add costs.

存量的既有环境（brownfield）也可以改造为 POL，但在并非为光纤设计过的空间里布放光缆，往往会增加成本。

From a vertical perspective, education institutions, hotels, hospitals and large corporate campuses are all good candidates for POL.

从行业角度看，**教育机构、酒店、医院和大型企业园区**都是 POL 的良好候选。

Even if building retrofitting isn't an option, intra-building and broad campus use cases for POL are compelling.

即便建筑改造不可行，楼内和广泛院区场景下的 POL 应用也很有吸引力。

Customers from various industries value different merits from POL.

不同行业的客户看重 POL 的不同优点。

For instance, on university campuses, there challenging environments, including dormitories, offices, libraries, canteens and labs.

例如在大学校园里，宿舍、办公室、图书馆、食堂和实验室都是很有挑战的环境；

With POL, all services can be bundled into one.

有了 POL，所有业务可以合并到一张网上。

In hospitality, cable television can be carried as well as data and phone services.

在酒店业，除了数据和电话业务，还可以承载有线电视。

POL is efficient in space and energy usage, making it attractive to large campuses and buildings.

POL 在空间和能耗上都更高效，这使其对大型院区和建筑很有吸引力。

Companies should investigate POL as an alternative to traditional designs as a way to introduce long term savings for increasing bandwidth needs, greater operational efficiencies, and lower operating costs.

企业应把 POL 作为传统设计的替代方案去调研，用它来实现长期节省——应对不断增长的带宽需求、更高的运营效率和更低的运营成本。

They should not, however, use any POL system that isn't managed with the same tools that their primary network is managed with.

但是，**他们不应采用任何无法用其主网络相同工具进行管理的 POL 系统。**

关键术语中英对照 / Key Terms

POL (Passive Optical LAN)	无源光局域网：PON 技术在企业/院区局域网场景的落地形态
PON (Passive Optical Network)	无源光网络：点到多点（P2MP）的光接入网
Passive	无源：中间分路器不供电；只有发送端与接收端需供电
Optical splitter	光分路器/分光器：把一路光均分为 N 路的无源器件
Single mode fiber	单模光纤
Pluggable optical transceivers	可插拔光模块
Backhaul	回传（通常指接入到汇聚/核心的上行链路）
Brownfield	存量/既有环境（相对于新建 greenfield）
Single pane of glass	单一管理界面：一套网管统管全网

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